

HCSM-27

Klein Orbit Structure

The Klein Four Group Action, Kernel Classification,
and the Framework Constant Four Orbit

The Hodge Complex Standard Model

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Abstract

The Hodge Complex Standard Model assigns the fourteen modes at the self paired eigenvalue of the discrete Laplacian on the periodic 8×8 torus to one Standard Model generation. The assignment of each mode to a specific Standard Model particle, together with the $N/4$ mass label attached to each, is the framework's *empirical thesis* (HCSM-23), structurally parallel to the Standard Model's Yukawa-coupling sector. The substrate derives the fourteen modes, their coordinates, the D_4 point group action, and the operations that act on the multiplet.

Two order-two operations act naturally on this multiplet: the diagonal reflection rs , identified in HCSM-26 as the unique chiral reflection, and the half period translation T_{half} . The two commute and each squares to the identity, so they generate a Klein four group. This paper proves four results about the action of this group. The Klein Orbit Theorem (derived), the Kernel Classification Theorem (derived), and the Swap Pair Bridge Theorem (derived at coordinate level, with the naming of the pair applied from HCSM-23) are theorems of the substrate and use no $N/4$ anchors. The Four-Orbit Framework-Constant Theorem is a theorem *given* the empirical identification: it states that the four-cycle orbit's four anchored $N/4$ values are simultaneously framework constants, satisfy two additive identities, and sum to fifty-six, the dimension of the Hodge sector at the self paired eigenvalue; their mean equals fourteen, the multiplicity. Section 8 catalogs which results are derived and which are anchored. All numerical claims are verified at machine precision.

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1 Introduction

1.1 The Klein group in HCSM

The Hodge Complex Standard Model (HCSM) derives the gauge structure, matter content, particle spectrum, and mixing matrices of the Standard Model from a single discrete substrate: the periodic 8×8 torus [1]. The fourteen modes at the self-paired eigenvalue $\lambda = 4$ of the discrete Laplacian are identified with one Standard Model generation. Their D_4 representation theory determines the gauge algebra; their $N/4$ labels determine the mass spectrum; the zero-state vector from the 14×19 measurement matrix determines the flavor structure.

The identification of the fourteen modes with the fourteen Standard Model particles, and the $N/4$ mass label attached to each, is the framework's *empirical thesis* (HCSM-23 [7]). It is not a derived result. The $N/4$ labels are empirical anchors, structurally parallel to the Standard Model's Yukawa couplings.

Throughout this paper:

- The fourteen modes, their coordinates, the D_4 point-group action, the diagonal reflection rs , the half-period translation T_{half} , and the Klein four-group V they generate are *derived* from the substrate.
- The identification of each mode with a specific Standard Model particle, and the $N/4$ label attached to each, are *empirical anchors* (HCSM-23).
- The structural relations among the $N/4$ values are *theorems given the anchor*.

Two order two operations act on the fourteen mode multiplet. The first is the diagonal reflection rs , which swaps the two mode coordinates and has been identified in HCSM-26 [8] as the unique chiral reflection of the D_4 point group. The second is the half-period translation T_{half} , which shifts both mode coordinates by half the torus side length. The two operations commute, and each squares to the identity, so they generate a Klein four group.

This paper proves that the action of the Klein group organizes the fourteen mode multiplet completely: the orbits are classified, the stabilizers of the orbits are classified, and one orbit in particular the four-cycle orbit contains four $N/4$ anchors whose values are all framework constants, under the empirical identification.

1.2 The one line summary

The Klein four group $V = \langle rs, T_{\text{half}} \rangle$ acts on the fourteen mode multiplet with five size-2 orbits and one size-4 orbit. The pointwise stabilizers of these orbits fall into the three non-trivial subgroups of the Klein group. One orbit the four-cycle orbit is the framework-constant orbit: given the empirical identification, its four $N/4$ anchors are simultaneously framework constants and their sum equals 56, the dimension of the Hodge sector at the self-paired eigenvalue.

1.3 Derivation status of the results in this paper

The paper proves four results. They split into three groups according to their derivation status.

1.3.1 Fully derived (no anchor)

These results use only substrate primitives and do not refer to the $N/4$ anchors or to the particle identifications:

- (1) **Klein Orbit Theorem** (Theorem 2). The Klein four-group $V = \langle rs, T_{\text{half}} \rangle$ acts on the fourteen-mode multiplet with orbit decomposition consisting of five size-2 orbits and one size-4 orbit.
- (2) **Kernel Classification Theorem** (Theorem 5). The pointwise stabilizer of each Klein orbit falls into one of the three non-trivial subgroups of the Klein group: four orbits with stabilizer $\{e, rs \cdot T_{\text{half}}\}$, one with $\{e, rs\}$, and one with $\{e\}$.
- (3) **Swap-Pair Bridge Theorem** (Theorem 9). At coordinate level, the edge $\{(5, 7), (6, 6)\}$ is the unique edge in the distinguished-pair graph of the fourteen-mode multiplet that connects two distinct Klein orbits. Under the empirical identification, the edge is $\{\text{dark}, t\}$.

1.3.2 Theorem given the empirical identification

This result is conditional on the identification thesis (HCSM-23). It is a theorem about the structural relations among the $N/4$ anchors:

- (4) **Four-Orbit Framework-Constant Theorem** (Theorem 11). Given the empirical identification, the four-cycle orbit's four $N/4$ anchors are simultaneously framework constants: $(d-1)^2$, $(d-1)d$, the multiplicity, and $(d-1)(2d-1)$. They satisfy two additive identities within the orbit, and their sum equals $2(d-1)(2d-1) + 14 = 56$, the dimension of the Hodge sector at the self-paired eigenvalue. Their mean equals 14, the multiplicity.

1.4 Cross references to the HCSM paper stack

The paper depends on the following companion papers:

- HCSM-04 [3]: the fourteen-mode multiplet and its D_4 orbit decomposition.
- HCSM-13 [4]: the processor T and the F/B -class correspondence.
- HCSM-14 [5]: the winding generator spectrum and class partition.
- HCSM-22 [6]: the zero state. Referenced for the standard mode order.
- HCSM-23 [7]: the identification thesis and the $N/4$ labels.
- HCSM-26 [8]: the observer pair. Referenced for the diagonal reflection, the observer pair definition, and the Fixed Locus Theorem.

1.5 Organization

Section 2 states the setup: the fourteen-mode multiplet, the diagonal reflection, the half-period translation, and the Klein structure. Section 3 proves the Klein Orbit Theorem. Section 4 proves the Kernel Classification Theorem. Section 5 proves the Swap-Pair Bridge Theorem. Section 6 proves the Four-Orbit Framework-Constant Theorem. Section 7 discusses the structural interpretation. Section 8 catalogs inputs and derived results. Section 9 gives reproducibility notes. Section 10 concludes.

2 Setup

2.1 The fourteen-mode multiplet

The substrate is the periodic eight-by-eight torus $\Lambda = \mathbb{Z}_8 \times \mathbb{Z}_8$ with $N = 64$ sites [1, 2]. The discrete five-point Laplacian has eigenvalues

$$\lambda(n_1, n_2) = 4 - 2 \cos\left(\frac{\pi n_1}{4}\right) - 2 \cos\left(\frac{\pi n_2}{4}\right), \quad (n_1, n_2) \in \{0, \dots, 7\}^2. \quad (1)$$

The self-paired eigenvalue $\lambda = 4$ has multiplicity fourteen. The fourteen modes at $\lambda = 4$ decompose under the D_4 point group of the square into three orbits (HCSM-04 [3]):

$$\mathcal{O}_0 = \{(0, 4), (4, 0)\}, \quad (2)$$

$$\mathcal{O}_1 = \{(1, 3), (1, 5), (3, 1), (3, 7), (5, 1), (5, 7), (7, 3), (7, 5)\}, \quad (3)$$

$$\mathcal{O}_2 = \{(2, 2), (2, 6), (6, 2), (6, 6)\}. \quad (4)$$

In the standard mode order (matching the zero-state vector of HCSM-22 [6]), the fourteen modes are

$$(\nu, e, u, d, s, \mu, \text{dark}, c, \tau, b, W, Z, H, t). \quad (5)$$

The mode coordinates in this order are:

$$\begin{array}{llll} \nu : (0, 4), & e : (4, 0), & u : (1, 3), & d : (1, 5), \\ s : (3, 1), & \mu : (3, 7), & \text{dark} : (5, 7), & c : (7, 3), \\ \tau : (7, 5), & b : (5, 1), & W : (2, 2), & Z : (2, 6), \\ H : (6, 2), & t : (6, 6). \end{array}$$

The particle names attached to these coordinates are the empirical thesis of HCSM-23 [7]. The coordinates are derived from the substrate; the naming is anchored.

Throughout this paper, the primary presentation of every theorem is at the coordinate level. The particle names appear where needed for readability, and the notation $\{\text{dark}, t\}$ denotes the coordinate pair $\{(5, 7), (6, 6)\}$ under the identification thesis.

2.2 The diagonal reflection

The diagonal reflection is the map

$$rs : (n_1, n_2) \mapsto (n_2, n_1) \pmod{8}. \quad (6)$$

It is an element of the D_4 point group of the square, and it satisfies $rs^2 = e$. It has been identified in HCSM-26 [8] as the unique chiral reflection of D_4 : the unique element that fixes exactly the coordinate pair $\{(2, 2), (6, 6)\}$ on the fourteen-mode multiplet and anticommutes with the winding generator.

The fixed set of rs on the fourteen-mode multiplet is exactly the coordinate pair $\{(2, 2), (6, 6)\}$ (HCSM-26 Lemma 3.1). Under the identification thesis, this pair is $\{W, t\}$.

2.3 The half-period translation

The half-period translation is the map

$$T_{\text{half}} : (n_1, n_2) \mapsto (n_1 + 4, n_2 + 4) \pmod{8}. \quad (7)$$

It shifts both mode coordinates by half the torus side length $L/2 = 4$. It satisfies $T_{\text{half}}^2 = e$.

2.4 The Klein four-group

Lemma 1 (Klein structure). *The diagonal reflection rs and the half-period translation T_{half} commute and each squares to the identity:*

$$rs^2 = e, \quad T_{\text{half}}^2 = e, \quad [rs, T_{\text{half}}] = 0. \quad (8)$$

Hence they generate a Klein four-group

$$V = \langle rs, T_{\text{half}} \rangle = \{e, rs, T_{\text{half}}, rs \cdot T_{\text{half}}\}. \quad (9)$$

Proof. The relations $rs^2 = e$ and $T_{\text{half}}^2 = e$ are immediate from the definitions. For the commutator: for any mode (n_1, n_2) ,

$$rs(T_{\text{half}}(n_1, n_2)) = rs(n_1 + 4, n_2 + 4) = (n_2 + 4, n_1 + 4), \quad (10)$$

$$T_{\text{half}}(rs(n_1, n_2)) = T_{\text{half}}(n_2, n_1) = (n_2 + 4, n_1 + 4). \quad (11)$$

The two expressions coincide. Hence $[rs, T_{\text{half}}] = 0$. The group generated by two commuting involutions is the Klein four-group; this is verified at machine precision in the companion verification script. $\square$

2.5 The product $rs \cdot T_{\text{half}}$

The product of the two generators is the map

$$rs \cdot T_{\text{half}} : (n_1, n_2) \mapsto (n_2 + 4, n_1 + 4) \pmod{8}. \quad (12)$$

It is an involution, and it is not equal to either rs or T_{half} .

2.6 The $N/4$ anchors

The $N/4$ values are the framework's empirical anchors for the particle spectrum (HCSM-23 [7]). In the standard mode order:

$$(N/4)_i = (60, 24, 21, 20, 14, 14, 12, 9.5, 9, 7.5, 2, 2, 1, 0.5). \quad (13)$$

The values are treated as inputs throughout this paper. The structural relations among them are the theorems. Throughout, we refer to these values using descriptive names; e.g., the dark-photon anchor equals twelve, and the top-quark anchor equals one half. The reader may consult the coordinate-level statement of each theorem and substitute the particle names for the corresponding coordinates.

3 The Klein Orbit Theorem

This section proves that the Klein group acts on the fourteen-mode multiplet with five size-2 orbits and one size-4 orbit. The result is derived at the coordinate level.

Orbit	Coordinates (particle names)	Size
V_1	$\{(0, 4), (4, 0)\} \quad (\{\nu, e\})$	2
V_2	$\{(6, 2), (2, 6)\} \quad (\{H, Z\})$	2
V_3	$\{(1, 5), (5, 1)\} \quad (\{d, b\})$	2
V_4	$\{(3, 7), (7, 3)\} \quad (\{\mu, c\})$	2
V_5	$\{(2, 2), (6, 6)\} \quad (\{W, t\})$	2
V_6	$\{(5, 7), (3, 1), (1, 3), (7, 5)\} \quad (\{\text{dark}, s, u, \tau\})$	4

Table 1: Klein orbit decomposition of the fourteen-mode multiplet. The orbit decomposition and the coordinate-level presentation are derived from the substrate. The particle names in parentheses are attached by the empirical identification (HCSM-23).

3.1 Orbit decomposition

Theorem 2 (Klein Orbit Theorem). *The Klein four-group $V = \langle rs, T_{\text{half}} \rangle$ acts on the fourteen-mode multiplet with orbit decomposition consisting of five orbits of size two and one orbit of size four. Explicitly, the orbits are given in Table 1.*

The total mode count is $5 \cdot 2 + 1 \cdot 4 = 14$, matching the multiplicity at the self-paired eigenvalue.

Proof. Direct computation of the orbit of each coordinate under the Klein group. We compute representative orbits.

Orbit of $(0, 4)$:

$$\begin{aligned}
 e \cdot (0, 4) &= (0, 4), \\
 rs \cdot (0, 4) &= (4, 0), \\
 T_{\text{half}} \cdot (0, 4) &= (4, 0), \\
 (rs \cdot T_{\text{half}}) \cdot (0, 4) &= (0, 4).
 \end{aligned}$$

The orbit is $\{(0, 4), (4, 0)\}$, size 2. This is V_1 .

Orbit of $(6, 2)$:

$$\begin{aligned}
 e \cdot (6, 2) &= (6, 2), \\
 rs \cdot (6, 2) &= (2, 6), \\
 T_{\text{half}} \cdot (6, 2) &= (2, 6), \\
 (rs \cdot T_{\text{half}}) \cdot (6, 2) &= (6, 2).
 \end{aligned}$$

The orbit is $\{(6, 2), (2, 6)\}$, size 2. This is V_2 .

Orbit of $(1, 5)$:

$$\begin{aligned}
 e \cdot (1, 5) &= (1, 5), \\
 rs \cdot (1, 5) &= (5, 1), \\
 T_{\text{half}} \cdot (1, 5) &= (5, 1), \\
 (rs \cdot T_{\text{half}}) \cdot (1, 5) &= (1, 5).
 \end{aligned}$$

The orbit is $\{(1, 5), (5, 1)\}$, size 2. This is V_3 .

Orbit of (3, 7):

$$\begin{aligned} e \cdot (3, 7) &= (3, 7), \\ rs \cdot (3, 7) &= (7, 3), \\ T_{\text{half}} \cdot (3, 7) &= (7, 3), \\ (rs \cdot T_{\text{half}}) \cdot (3, 7) &= (3, 7). \end{aligned}$$

The orbit is $\{(3, 7), (7, 3)\}$, size 2. This is V_4 .

Orbit of (2, 2):

$$\begin{aligned} e \cdot (2, 2) &= (2, 2), \\ rs \cdot (2, 2) &= (2, 2), \\ T_{\text{half}} \cdot (2, 2) &= (6, 6), \\ (rs \cdot T_{\text{half}}) \cdot (2, 2) &= (6, 6). \end{aligned}$$

The orbit is $\{(2, 2), (6, 6)\}$, size 2. This is V_5 .

Orbit of (5, 7):

$$\begin{aligned} e \cdot (5, 7) &= (5, 7), \\ rs \cdot (5, 7) &= (7, 5), \\ T_{\text{half}} \cdot (5, 7) &= (1, 3), \\ (rs \cdot T_{\text{half}}) \cdot (5, 7) &= (3, 1). \end{aligned}$$

The orbit is $\{(5, 7), (7, 5), (1, 3), (3, 1)\}$, size 4. This is V_6 .

The six orbits cover all fourteen coordinates. The orbit decomposition is as stated. $\square$

3.2 The three structural regions

The orbit decomposition partitions the fourteen-mode multiplet into three structurally distinct regions:

The four two-cycle orbits V_1 – V_4 . Each of these orbits has two members that are exchanged by the diagonal reflection and by the half-period translation, and that are fixed by the product $rs \cdot T_{\text{half}}$. They are the framework's standard mirror pairs.

The observer orbit V_5 . This orbit is $\{(2, 2), (6, 6)\}$. Both members are fixed by the diagonal reflection rs . Under the identification thesis, this is the observer pair of HCSM-26 [8].

The four-cycle orbit V_6 . This orbit contains four coordinates that are cyclically exchanged by the Klein group. It contains the swap partner of the top-quark coordinate and has the framework-constant property of Section 6.

Remark 3. The four-cycle orbit V_6 contains the coordinates $(5, 7)$, $(7, 5)$, $(1, 3)$, $(3, 1)$, all drawn from the middle D_4 orbit $\mathcal{O}_1$. The Klein group therefore further refines $\mathcal{O}_1$: the D_4 orbit of size 8 splits into the three Klein orbits $V_3 = \{(1, 5), (5, 1)\}$, $V_4 = \{(3, 7), (7, 3)\}$, and $V_6 = \{(5, 7), (3, 1), (1, 3), (7, 5)\}$.

4 Kernel Classification

This section classifies the pointwise stabilizer of each Klein orbit. The result is derived at the coordinate level.

4.1 Definition

Definition 4 (Pointwise stabilizer). For a Klein orbit $V_i \subset \mathcal{M}$, the pointwise stabilizer is the subgroup of V consisting of elements g such that $g \cdot m = m$ for every mode $m \in V_i$.

4.2 Classification

Theorem 5 (Kernel Classification Theorem). *The pointwise stabilizer of each Klein orbit is one of the three non-trivial subgroups of the Klein group. The classification is given in Table 2.*

Orbit	Members (coordinate-level)	Pointwise stabilizer
V_1	$\{(0, 4), (4, 0)\}$	$\{e, rs \cdot T_{\text{half}}\}$
V_2	$\{(6, 2), (2, 6)\}$	$\{e, rs \cdot T_{\text{half}}\}$
V_3	$\{(1, 5), (5, 1)\}$	$\{e, rs \cdot T_{\text{half}}\}$
V_4	$\{(3, 7), (7, 3)\}$	$\{e, rs \cdot T_{\text{half}}\}$
V_5	$\{(2, 2), (6, 6)\}$	$\{e, rs\}$
V_6	$\{(5, 7), (3, 1), (1, 3), (7, 5)\}$	$\{e\}$

Table 2: Pointwise stabilizer classification of the six Klein orbits.

Proof. Direct computation of the action of each group element on each orbit. We compute representative cases.

Orbit $V_1 = \{(0, 4), (4, 0)\}$:

$$\begin{aligned}
 rs \cdot (0, 4) &= (4, 0) \neq (0, 4) && \text{(swaps),} \\
 T_{\text{half}} \cdot (0, 4) &= (4, 0) \neq (0, 4) && \text{(swaps),} \\
 (rs \cdot T_{\text{half}}) \cdot (0, 4) &= (0, 4), \quad (rs \cdot T_{\text{half}}) \cdot (4, 0) = (4, 0) && \text{(fixed pointwise).}
 \end{aligned}$$

Hence the pointwise stabilizer is $\{e, rs \cdot T_{\text{half}}\}$.

Orbit $V_5 = \{(2, 2), (6, 6)\}$:

$$\begin{aligned}
 rs \cdot (2, 2) &= (2, 2), \quad rs \cdot (6, 6) = (6, 6) && \text{(fixed pointwise),} \\
 T_{\text{half}} \cdot (2, 2) &= (6, 6) \neq (2, 2) && \text{(swaps),} \\
 (rs \cdot T_{\text{half}}) \cdot (2, 2) &= (6, 6) \neq (2, 2) && \text{(swaps).}
 \end{aligned}$$

Hence the pointwise stabilizer is $\{e, rs\}$.

Orbit $V_6 = \{(5, 7), (3, 1), (1, 3), (7, 5)\}$:

$$\begin{aligned}
 rs \cdot (5, 7) &= (7, 5) \neq (5, 7) && \text{(swaps),} \\
 T_{\text{half}} \cdot (5, 7) &= (1, 3) \neq (5, 7) && \text{(swaps),} \\
 (rs \cdot T_{\text{half}}) \cdot (5, 7) &= (3, 1) \neq (5, 7) && \text{(swaps).}
 \end{aligned}$$

No non-identity element fixes every member of V_6 . Hence the pointwise stabilizer is $\{e\}$.

The remaining orbits are computed analogously and yield the classification as stated. $\square$

4.3 The three stabilizer classes

Corollary 6 (Three stabilizer classes). *The Klein orbits partition into three stabilizer classes:*

- Class $\{e, rs \cdot T_{\text{half}}\}$: four orbits, V_1, V_2, V_3, V_4 .
- Class $\{e, rs\}$: one orbit, V_5 .
- Class $\{e\}$: one orbit, V_6 .

Proof. Immediate from Theorem 5. □

Remark 7. The observer orbit V_5 is the unique orbit whose pointwise stabilizer is generated by the chiral reflection rs . This is consistent with the Chirality Selection Theorem of HCSM-26 [8]: the coordinate pair $\{(2, 2), (6, 6)\}$ is the unique fixed locus of rs on the fourteen-mode multiplet.

5 The Swap-Pair Bridge Theorem

This section proves that the orbit-class swap pair is the unique edge in the distinguished-pair graph that connects two distinct Klein orbits. The graph structure and the edge identification are derived at the coordinate level. The naming of the swap pair as $\{\text{dark}, t\}$ applies the empirical identification (HCSM-23).

5.1 The distinguished-pair graph

Definition 8 (Distinguished-pair graph). The distinguished-pair graph of the fourteen-mode multiplet has a vertex for each mode and an edge for each two-element subset that is distinguished by at least one of the following structural criteria: (i) being fixed by an involution of the D_4 point group; (ii) being a two-cycle of the diagonal reflection rs ; (iii) being a two-cycle of the half-period translation T_{half} ; and (iv) being the orbit-class swap pair (defined in HCSM-26 [8]).

The distinguished-pair graph has ten edges, listed at coordinate level in Table 3. Its vertex set is a subset of the fourteen coordinates.

Index	Edge (coordinates; names)
1	$\{(0, 4), (4, 0)\} (\{\nu, e\})$
2	$\{(1, 3), (3, 1)\} (\{u, s\})$
3	$\{(1, 3), (5, 7)\} (\{u, \text{dark}\})$
4	$\{(1, 5), (5, 1)\} (\{d, b\})$
5	$\{(3, 7), (7, 3)\} (\{\mu, c\})$
6	$\{(3, 1), (7, 5)\} (\{s, \tau\})$
7	$\{(5, 7), (7, 5)\} (\{\text{dark}, \tau\})$
8	$\{(5, 7), (6, 6)\} (\{\text{dark}, t\})$
9	$\{(2, 6), (6, 2)\} (\{Z, H\})$
10	$\{(2, 2), (6, 6)\} (\{W, t\})$

Table 3: The ten edges of the distinguished-pair graph.

5.2 The inter-orbit edges

Theorem 9 (Swap-Pair Bridge Theorem). *At coordinate level, the edge $\{(5, 7), (6, 6)\}$ is the unique edge in the distinguished-pair graph of the fourteen-mode multiplet that connects two distinct Klein orbits. Under the empirical identification (HCSM-23), the edge is $\{\text{dark}, t\}$.*

Proof. The ten edges of the distinguished-pair graph are listed in Table 3.

Within-orbit edges. The following nine edges have both endpoints in the same Klein orbit:

$$\begin{aligned} \{(1, 3), (3, 1)\}, \{(1, 3), (5, 7)\}, \{(3, 1), (7, 5)\}, \{(5, 7), (7, 5)\} &\subset V_6, \\ \{(0, 4), (4, 0)\} &\subset V_1, \\ \{(1, 5), (5, 1)\} &\subset V_3, \\ \{(3, 7), (7, 3)\} &\subset V_4, \\ \{(2, 6), (6, 2)\} &\subset V_2, \\ \{(2, 2), (6, 6)\} &\subset V_5. \end{aligned}$$

The remaining edge. The edge $\{(5, 7), (6, 6)\}$ has its two endpoints in distinct Klein orbits:

$$(5, 7) \in V_6 = \{(5, 7), (3, 1), (1, 3), (7, 5)\}, \quad (6, 6) \in V_5 = \{(2, 2), (6, 6)\}. \quad (14)$$

Hence $\{(5, 7), (6, 6)\}$ is the unique inter-orbit edge. $\square$

Corollary 10 (Bridge structure). *The edge $\{(5, 7), (6, 6)\}$ is the unique edge of the distinguished-pair graph that bridges the four-cycle orbit V_6 and the observer orbit V_5 . It is the only connection between the four-cycle orbit and the observer orbit. Under the empirical identification, this edge is the swap pair $\{\text{dark}, t\}$.*

Proof. Immediate from Theorem 9. $\square$

6 The Four-Orbit Framework-Constant Theorem

This section proves that, given the empirical identification (HCSM-23), the four-cycle orbit's four $N/4$ anchors are simultaneously framework constants.

6.1 The four-cycle orbit's anchored values

Given the empirical identification (HCSM-23), the four coordinates of the four-cycle orbit V_6 carry the following $N/4$ anchors:

$$N/4(\tau) = 9, \quad (15)$$

$$N/4(\text{dark}) = 12, \quad (16)$$

$$N/4(s) = 14, \quad (17)$$

$$N/4(u) = 21. \quad (18)$$

At the coordinate level, these anchors attach to the four coordinates $(7, 5)$, $(5, 7)$, $(3, 1)$, $(1, 3)$ respectively.

6.2 The theorem

Theorem 11 (Four-Orbit Framework-Constant Theorem). *Given the empirical identification (HCSM-23), the four $N/4$ anchors of the four-cycle orbit V_6 are simultaneously simple framework constants:*

$$N/4(\tau) = (d - 1)^2 = 9, \quad (19)$$

$$N/4(\text{dark}) = (d - 1) \cdot d = 12, \quad (20)$$

$$N/4(s) = \text{multiplicity} = 14, \quad (21)$$

$$N/4(u) = (d - 1)(2d - 1) = 21. \quad (22)$$

They satisfy two additive identities within the orbit:

$$N/4(\tau) + N/4(\text{dark}) = N/4(u), \quad (23)$$

$$N/4(\tau) + N/4(s) = H + 2d - 1 = 23, \quad (24)$$

where $H = 2^d = 16$ is the hidden-sector dimension parameter. The sum of the four anchors equals

$$N/4(\tau) + N/4(\text{dark}) + N/4(s) + N/4(u) = 9 + 12 + 14 + 21 = 56, \quad (25)$$

which is the dimension of the Hodge sector at the self-paired eigenvalue. The mean of the four anchors equals

$$\frac{56}{4} = 14, \quad (26)$$

which is the multiplicity at the self-paired eigenvalue.

Proof. Direct arithmetic on the anchors. For the additive identities:

$$9 + 12 = 21 (= N/4(u)), \quad (27)$$

$$9 + 14 = 23 (= H + 2d - 1). \quad (28)$$

For the sum:

$$9 + 12 + 14 + 21 = 56 = 14 + 28 + 14 = \dim(\Omega^0 \oplus \Omega^1 \oplus \Omega^2)|_{\lambda=4}. \quad (29)$$

For the mean:

$$\frac{56}{4} = 14 = 2(L - 1) = \text{multiplicity}_{\lambda=4}. \quad (30)$$

□

6.3 The algebraic derivation of the sum

Proposition 12 (Algebraic derivation of the sum). *Given the empirical identification (HCSM-23), the sum of the four-cycle orbit's $N/4$ anchors has the framework-constant form*

$$N/4(\tau) + N/4(\text{dark}) + N/4(s) + N/4(u) = 2(d - 1)(2d - 1) + 14. \quad (31)$$

At $d = 4$, this evaluates to $2 \cdot 3 \cdot 7 + 14 = 42 + 14 = 56$.

Proof. Add the four framework-constant forms:

$$N/4(\tau) + N/4(\text{dark}) + N/4(s) + N/4(u) = (d-1)^2 + (d-1)d + 14 + (d-1)(2d-1). \quad (32)$$

Collect the terms in $(d-1)$:

$$= (d-1)[(d-1) + d + (2d-1)] + 14 \quad (33)$$

$$= (d-1)[4d-2] + 14 \quad (34)$$

$$= 2(d-1)(2d-1) + 14. \quad (35)$$

At $d = 4$: $2 \cdot 3 \cdot 7 + 14 = 42 + 14 = 56$. $\square$

Corollary 13 (Four-orbit as a shadow of the Hodge sector). *Given the empirical identification (HCSM-23), the four-cycle orbit is a four-element shadow of the 56-dimensional Hodge sector at the self-paired eigenvalue: its four $N/4$ anchors sum to the sector dimension, and their mean is the multiplicity.*

Proof. Immediate from Theorem 11. $\square$

7 Discussion

7.1 What the Klein structure gives

The Klein four-group $V = \langle rs, T_{\text{half}} \rangle$ organizes the fourteen-mode multiplet into six orbits at coordinate level. The orbit decomposition and its stabilizer classification are derived from the substrate. The six orbits are:

- Four two-cycle orbits with pointwise stabilizer $\{e, rs \cdot T_{\text{half}}\}$: $V_1 = \{(0, 4), (4, 0)\}$, $V_2 = \{(6, 2), (2, 6)\}$, $V_3 = \{(1, 5), (5, 1)\}$, $V_4 = \{(3, 7), (7, 3)\}$.
- One two-cycle orbit with pointwise stabilizer $\{e, rs\}$: the observer orbit $V_5 = \{(2, 2), (6, 6)\}$.
- One four-cycle orbit with trivial pointwise stabilizer: $V_6 = \{(5, 7), (3, 1), (1, 3), (7, 5)\}$.

The edge $\{(5, 7), (6, 6)\}$ (the swap pair $\{\text{dark}, t\}$ under the empirical identification) is the unique bridge between the four-cycle orbit and the observer orbit. This is the framework's minimal connective structure between its two most distinctive orbits.

7.2 Refinement of the D_4 orbit partition

The D_4 orbit partition of the fourteen-mode multiplet is

$$\mathcal{O}_0 = \{(0, 4), (4, 0)\}, \quad \mathcal{O}_1 = \{(1, 3), (1, 5), (3, 1), (3, 7), (5, 1), (5, 7), (7, 3), (7, 5)\}, \quad (36)$$

$$\mathcal{O}_2 = \{(2, 2), (2, 6), (6, 2), (6, 6)\}. \quad (37)$$

The Klein orbit partition refines the D_4 orbit partition:

- $\mathcal{O}_0$ is the single Klein orbit $V_1 = \{(0, 4), (4, 0)\}$.
- $\mathcal{O}_1$ decomposes into three Klein orbits: $V_3 = \{(1, 5), (5, 1)\}$, $V_4 = \{(3, 7), (7, 3)\}$, and $V_6 = \{(5, 7), (3, 1), (1, 3), (7, 5)\}$.
- $\mathcal{O}_2$ decomposes into two Klein orbits: $V_2 = \{(6, 2), (2, 6)\}$ and $V_5 = \{(2, 2), (6, 6)\}$.

The Klein group therefore provides a fine-grained partition of the multiplet, in which each D_4 orbit of size greater than two is further decomposed.

7.3 Relation to the observer pair

The coordinate pair $\{(2, 2), (6, 6)\}$ of HCSM-26 is the Klein orbit V_5 . Its pointwise stabilizer is $\{e, rs\}$, the subgroup generated by the chiral reflection. This is consistent with the Chirality Selection Theorem of HCSM-26: the coordinate pair is the unique fixed locus of rs on the fourteen-mode multiplet. Under the identification thesis, this pair is the observer pair $\{W, t\}$.

The four-cycle orbit V_6 has trivial pointwise stabilizer. Its swap partner with the observer orbit is the coordinate $(6, 6)$ (the top-quark coordinate). The two most distinctive orbits of the Klein structure are connected by the edge $\{(5, 7), (6, 6)\}$.

8 Inputs and derived

8.1 Derived from the substrate

The following results are proved without reference to the $N/4$ anchors or to the particle identifications:

- (1) The Klein structure $rs^2 = T_{\text{half}}^2 = e$ and $[rs, T_{\text{half}}] = 0$ (Lemma 1).
- (2) The orbit decomposition into five size-2 orbits and one size-4 orbit (Theorem 2).
- (3) The pointwise stabilizer classification of each orbit (Theorem 5 and Corollary 6).
- (4) The ten edges of the distinguished-pair graph and the unique inter-orbit edge $\{(5, 7), (6, 6)\}$ (Theorem 9 and Corollary 10).

8.2 Theorems given the empirical identification (HCSM-23)

The following result is a theorem about the structural relations among the $N/4$ anchors:

- (5) The Four-Orbit Framework-Constant Theorem (Theorem 11): the four $N/4$ anchors of the four-cycle orbit are simultaneously framework constants, satisfy two additive identities, and sum to 56.
- (6) The algebraic derivation of the sum (Proposition 12).
- (7) The four-orbit-as-shadow statement (Corollary 13).

8.3 Inputs

- (1) **Input 1 (SM content).** The Standard Model particles and their observed properties, as used to identify the fourteen modes.
- (2) **Input 2 (physical identifications).** The identifications $A_1 \leftrightarrow \text{color}$, $B_1 \leftrightarrow \text{weak isospin}$.
- (3) **Input 3 (the identification thesis, HCSM-23).** The assignment of each of the fourteen modes at $\lambda = 4$ to a specific Standard Model particle, together with the $N/4$ mass label attached to each. This is the framework's empirical anchor, structurally parallel to the Standard Model's Yukawa couplings.
- (4) **The framework constants** $d = 4$, $N = 64$, $L = 8$, $H = 16$ (HCSM-01 [2]).

8.4 What is not claimed

We do not claim to derive the identification thesis (HCSM-23) from the substrate. The assignment of the fourteen modes to the fourteen particles, and the $N/4$ labels attached to each, are the framework’s empirical anchors.

We do not claim that the Four-Orbit Framework-Constant Theorem is HCSM-native without qualification. It is a theorem given the anchor: it holds for the specific anchor values observed in the Standard Model, and it is falsifiable if the anchor values differ.

We do not claim that the naming of the coordinates $(5, 7)$ and $(6, 6)$ as dark and t is derived. The coordinates and the edge structure are derived; the naming is anchored.

The framework’s claim is that it has fewer empirical anchors than the Standard Model, not that it has none.

9 Reproducibility

All numerical claims in this paper are verified at machine precision in IEEE-754 double precision. The verification script is provided in the HCSM companion repository at `12-VERIFICATION/HCSM-27_verification`. The script constructs the fourteen-mode multiplet, the diagonal reflection rs , the half-period translation T_{half} , the Klein four-group $V = \langle rs, T_{\text{half}} \rangle$, the orbit decomposition, the pointwise stabilizers, the distinguished-pair graph, and the four-cycle orbit’s anchors from HCSM-native definitions, and verifies:

Derived at coordinate level:

- (1) The Klein four-group presentation $rs^2 = T_{\text{half}}^2 = e$ and $[rs, T_{\text{half}}] = 0$.
- (2) The orbit decomposition into four two-cycle orbits with stabilizer $\{e, rs \cdot T_{\text{half}}\}$, one two-cycle orbit with stabilizer $\{e, rs\}$, and one four-cycle orbit with trivial stabilizer.
- (3) The pointwise stabilizer classification of each orbit, computed purely from coordinates.
- (4) The unique inter-orbit edge $\{(5, 7), (6, 6)\}$ in the distinguished-pair graph.

Given the empirical identification (HCSM-23):

- (5) The four-cycle orbit’s $N/4$ anchors and their framework-constant forms.
- (6) The additive identities $N/4(\tau) + N/4(\text{dark}) = N/4(u)$ and $N/4(\tau) + N/4(s) = H + 2d - 1$.
- (7) The sum = 56 and the mean = 14.

Running `12-VERIFICATION/run_all.py` reproduces every numerical claim in the HCSM paper stack.

10 Conclusion

This paper has characterized the Klein orbit structure of the fourteen-mode multiplet in the Hodge Complex Standard Model.

Three theorems are fully derived from the substrate and do not use the $N/4$ anchors. The Klein Orbit Theorem establishes that the Klein four-group generated by the diagonal reflection and the half-period translation acts on the multiplet with orbit decomposition consisting of five size-2 orbits and one size-4 orbit. The Kernel Classification Theorem establishes that the pointwise stabilizer of

each orbit is one of the three non-trivial subgroups of the Klein group: four orbits have stabilizer $\{e, rs \cdot T_{\text{half}}\}$, the observer orbit has stabilizer $\{e, rs\}$, and the four-cycle orbit has trivial stabilizer. The Swap-Pair Bridge Theorem establishes that the edge $\{(5, 7), (6, 6)\}$ is the unique inter-orbit edge of the distinguished-pair graph; under the empirical identification, this edge is $\{\text{dark}, t\}$.

One theorem is conditional on the empirical identification thesis (HCSM-23). Given the anchor, the four-cycle orbit's four $N/4$ values are simultaneously framework constants, satisfy two additive identities within the orbit, and sum to fifty-six, the dimension of the Hodge sector at the self-paired eigenvalue; their mean equals fourteen, the multiplicity. This theorem is falsifiable: it holds for the specific anchor values observed in the Standard Model.

The Klein structure organizes the fourteen-mode multiplet in a way that refines the D_4 orbit partition and brings the framework-constant arithmetic of the four-cycle orbit into focus. The observer pair of HCSM-26 is the orbit V_5 of this structure, and the framework-constant orbit is V_6 .

The framework's claim is that the Standard Model is what the substrate looks like when you observe it. The identification of the fourteen modes with the fourteen particles is the empirical thesis; the structural relations among the $N/4$ anchors are theorems given the thesis. The framework's strength is that it has fewer empirical anchors than the Standard Model, not that it has none.

References

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- [8] S. Preschutti, *HCSM-26: The Observer Pair $\{W, t\}$* , HCSM White Paper Series (2026).

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    are simultaneously framework constants and sum to fifty-six, the
    dimension of the Hodge sector at the self-paired eigenvalue.}
}
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